DATA VORTEX | ROUND 2 EVALUATION METRICS REPORT

Participant details:

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Team Size: 1

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1. Dataset & Experimental setup

The dataset contains 9,000 labelled social media text records. A group-aware split produced 7,183 training samples and 1,817 test samples. TF-IDF was fitted only on training data.

Component	Configuration
Dataset size	9,000
Training set	7,183
Test set	1,817
Sentiment classes	Negative, Neutral, Positive
TF-IDF	Unigrams + Bigrams
Features	25,650

Raw dataset

↓

Text cleaning

↓

Duplicate-aware group split

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Training / test data

↓

TF-IDF

↓

Model Training

↓

Evaluation

2. Model Comparison

Model	Accuracy	Weighted F1
Logistic Regression	58.83%	58.88%
Linear SVM	57.57%	57.62%
Multinomial Naive Bayes	59.49%	59.25%

Multinomial Naive Bayes produced the highest measured accuracy and weighted F1-score among the three evaluated configurations.

3. Final Model Evaluation

Multinomial Naive Bayes was selected as the final model because it achieved the highest measured accuracy and weighted F1 among the three evaluated configurations under identical experimental conditions. The metrics below are computed on the 1,817 held-out test samples

Metric	Score
Accuracy	59.49%
Macro Precision	59.34%
Macro Recall	59.40%
Weighted F1	59.25%
Macro F1	59.16%
Weighted Precision	59.43%
Weighted Recall	59.49%

4. Class-wise Performance

Class	Precision	Recall	F1-score	Support
Negative	65.59%	62.32%	63.91%	621
Neutral	53.83%	47.39%	50.40%	593
Positive	58.58%	68.49%	63.15%	603
Macro average	59.34%	59.40%	59.16%	1,817

Weighted average	59.43%	59.49%	59.25%	1,817

Observations

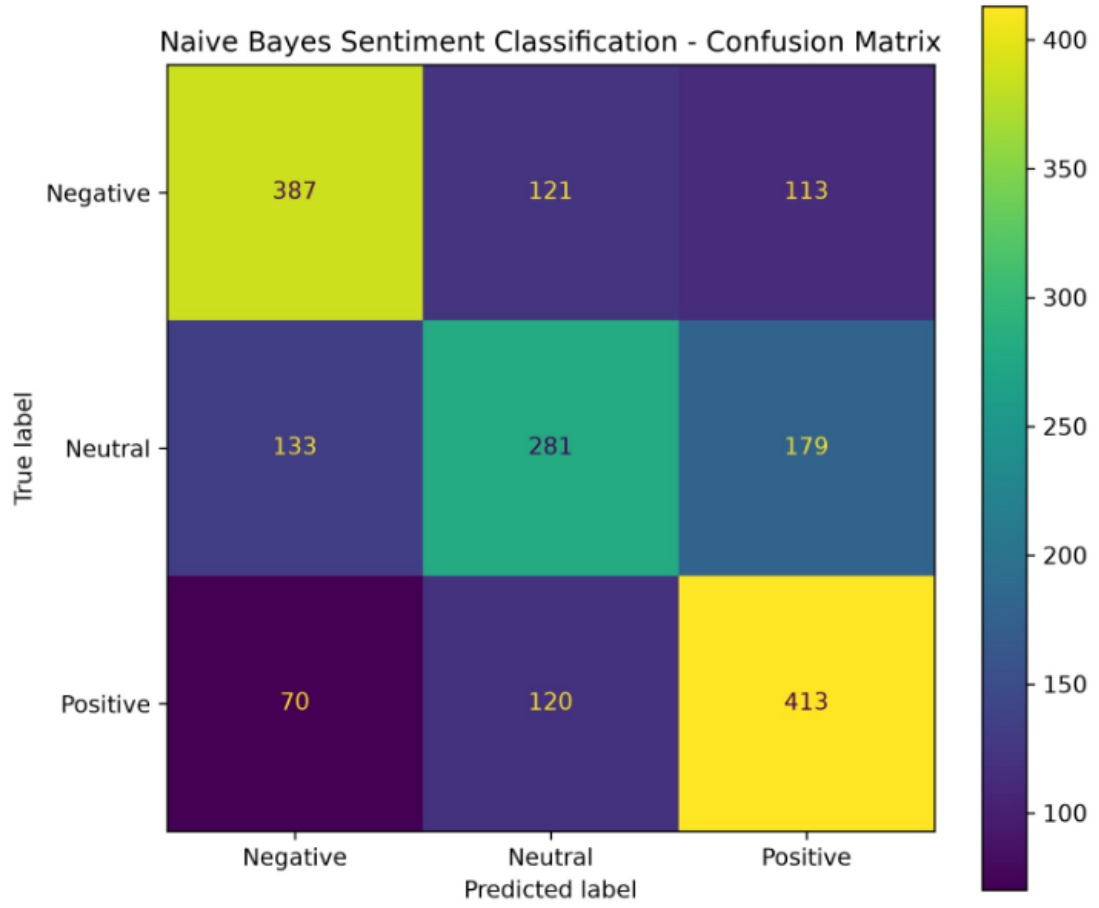
• **Negative** has the highest precision of the three classes (65.59%). Predictions of Negative are correct about two times in three, though recall of 62.32% means roughly 37% of genuinely negative posts are missed.

• **Positive** shows the widest precision–recall gap: recall is the highest of any class (68.49%) while precision is lower (58.58%). The model applies the Positive label readily, capturing most true positives at the cost of false ones.

• **Neutral** is the weakest class on every measure. Recall of 47.39% means fewer than half of all genuinely neutral posts are recovered, and its F1 of 50.40% sits 13.51 points below Negative.

5. Confusion Matrix

	Pred. Negative	Pred. Neutral	Pred. Positive	Row total
Actual Negative	387	121	113	621
Actual Neutral	133	281	179	593
Actual Positive	70	120	413	603
Column total	590	522	705	1,817



Correct predictions :

Negative → Negative = 387

Neutral → Neutral = 281

Positive → Positive = 413

Largest confusion: Neutral → Positive = 179

Second-largest confusion: Neutral → Negative = 133

The largest individual error pattern was Neutral posts being classified as Positive, followed by Neutral posts being classified as Negative.

6. Error Analysis

Total misclassified samples: 736 out of 1,817 test samples.

Actual	Predicated	Errors	% of Errors
Neutral	Positive	179	24.32%
Neutral	Negative	133	18.07%
Negative	Neutral	121	16.44%
Positive	Neutral	120	16.30%
Negative	Positive	113	15.35%
Positive	Negative	70	9.51%
Total	----	736	100%

observations

1. Neutral posts account for the largest directional confusion patterns.

2. Neutral→Positive is the largest individual error category.

3. Negative→Positive and Positive→Negative errors occur less frequently than the major Neutral-related confusions.

4. The results suggest that simple TF-IDF lexical representation has limitations when sentiment depends on context, implicit meaning, mixed sentiment, or subtle wording.